

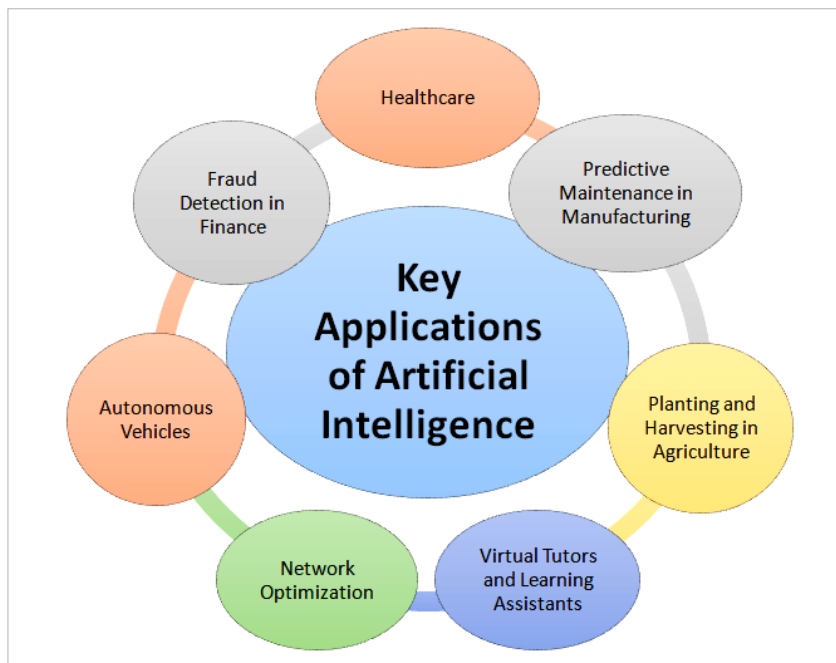


Figure 1: Key applications and use cases of AI

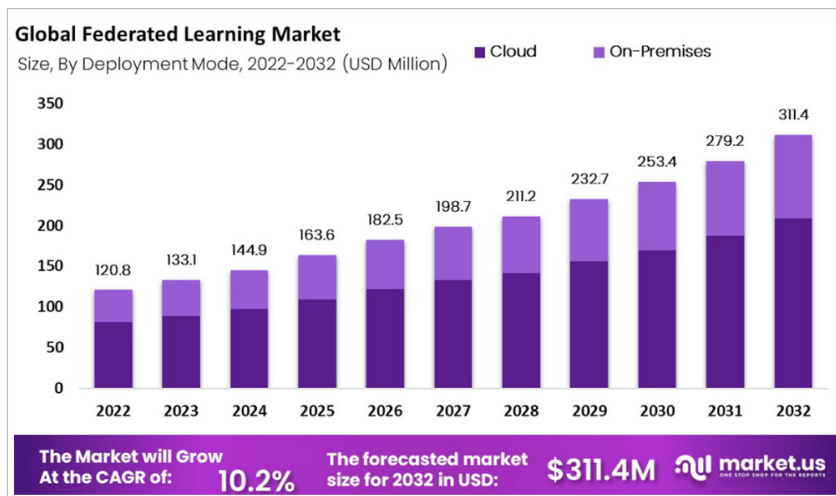


Figure 2: Market size of federated learning

- potential losses.
- **Risk management:** It can assess client portfolios and market conditions, helping firms manage risk more effectively.
- **Smart traffic management:** It optimises traffic patterns and signal timings, reducing road congestion and improving urban mobility.
- **Supply chain optimisation:** It can help predict and manage logistical challenges, enhancing efficiency right from warehouse operations to delivery routes.
- **Self-driving cars:** AI powers the software that allows vehicles to navigate and make decisions without human input, aiming to improve safety and reduce traffic incidents.

Automotive and transportation

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Retail

- **Customer experience:** AI algorithms offer personalised shopping experiences by recommending products based on browsing and purchase history.
- **Inventory management:** AI predicts market demand, helping retailers stock efficiently and reduce excess inventory.
- **Chatbots and virtual assistants:** These AI-powered tools handle customer enquiries instantaneously, improving service quality and operational efficiency.

Manufacturing

- **Predictive maintenance:** By monitoring equipment data, AI predicts when machines are likely to fail, scheduling maintenance to avoid costly downtime.
- **Quality assurance:** AI systems automate the inspection process for manufactured goods, enhancing product quality and consistency.
- **Production optimisation:** AI streamlines production processes, optimising resource use and output levels.

Education

- **Customised learning experiences:** AI adapts educational content to fit the learning speeds and styles of individual students, potentially improving educational outcomes.
- **Automation of routine tasks:** Tasks such as grading and attendance can be managed by AI, allowing educators to focus more on teaching.
- **Interactive learning tools:** AI-driven simulations and interactive modules enhance student engagement and facilitate active learning.

Media and entertainment

- **Content personalisation:** Streaming platforms use AI to analyse viewing habits and tailor content recommendations to individual preferences.